

This site has a total of 1,800 courses on offer from a diverse range of disciplines. Are you a student of Aeronautical Engineering who missed the lecture on Gusts and Winds? Were you too bored to attend that talk on Modal Metaphysics but need an idea of what on earth that meant? Just head over to the MIT Web site and do a search on the topic. All the courses are categorised into several broad areas like science, engineering, management.

Once you have got the course you were looking for, you can download the course materials as a zipped file. In most cases, the downloadable package will consist of text files. There are lecture notes, assignments, quizzes (along with answers), tutorials and details about reference books. The material is arranged in the form of a calendar, which gives you an idea of how the topics are arranged. Some modules also offer video or audio lectures to supplement the text. In fact, there are complete courses delivered in audio or video format. Such courses need software like QuickTime player for streaming video or Real Player for the downloadable files.

Apart from the classroom courses, the OCW Web site has other resources for maximising the impact of the material. These include training videos, instructors’ manuals, textbooks and online publications. For administrators and syllabus drafters, the MIT Curriculum Guide provides a glimpse of how MIT structures its courses, what are its requirements and what reference books are usually prescribed.

To ensure that the supply of quality students do not dry up, MIT has a section called “Highlights for High School” where students and teachers from the school level can find study material suitable for their level. There are videos of lab experiments, science projects, demonstrations and even courses on how to write better. The material is taken from MIT’s introductory courses and feature 11 subjects—from Computers and Electronics to Physical Education. High school courses developed by Educational Studies Program, a MIT student group established 50 years ago, are also listed on the site.

OCW has a forum as well, where users gather to discuss course materials. Since the Web site is updated regularly, users can subscribe to the sectional RSS feeds and also join the mailing list for accessing a regular newsletter. Links to archived courses, no longer available on the OCW site are also provided. All these courses are covered under a Creative Commons license, and hence can be used for free.

CMU’s Open Learning Initiative
(<http://www.cmu.edu/oli/>)

Carnegie Mellon University is another heavy-weight, which has an extensive and free e-learning program. You don’t even have to register to access the courses. However, these courses are used by some institutes to award credits and grades to their institutes. This means that the student will have to register for that particular course, and their performance statistics will be logged for supervision and oversight by their instructors. OLI courses are also used by students of CMU as part of their coursework.



An example of how MIT OpenCourseWare organises its courses

Similar to OCW, the courses as part of OLI also cover a wide range of subjects, though the selection is not as comprehensive. Currently, the site has courses in subjects like Engineering, Statistics, Chemistry, Logic and Proofs and Physics. Each course is divided into different modules, which usually comprise detailed lessons and courseware arranged topic-wise.

While a student can take courses without registration, he or she, can create a tracker, which will keep track of the course progress. Instructors who want to teach the OLI course have to register for an instructor account. After CMU completes a verification process, the instructor will receive additional privileges like access to student performance monitoring tools, expository text, case studies, simulations and other instructor support material.

CMU also conducts periodic evaluation studies to gauge the effectiveness of courses and keeps tweaking them. At the time of writing, the Web site was undergoing a major redesign after which the number of courses offered would increase and also become more extensive. It also conducts regular workshops and conferences for course development and evaluation in the university premises, which are open to all and can be joined for free. You would have to, however, pay for your own travel and stay.

OER Commons
(<http://www.oercommons.org/>)

This site is not affiliated to any university like the two above, but it features courses from various universities, colleges and training institutes. These courses are in different subjects and are available in different levels, from primary to post secondary. The broad areas covered are arts, humanities, business, mathematics and statistics, science and technology and social sciences. By its sheer volume, this Web site may be one of the largest of its kind on the Internet.

OER is different from the other e-learning resources in that users can share their own courses on the network, and also access courses that are already present. Submitted material has to go through a complex process of vetting and review before they are featured. So you could, in theory, find the physics lessons of your grade ten teacher who taught you that science was much more than mugging up formulae and boring facts. The resources on this site include lectures, demonstrations, full courses, modules, case studies, simulations, etc., in different media like video, podcasts, Flash movies and plain text files.

Despite the varied sources and diverse types of media on this site there is one commonality—everything is free and licensed under Creative Commons. That means users can download, use and modify any of these resources without the fear of getting sued for copyright infringement.

Users who want to contribute content to the community can create an account and log in to their portfolio. The community consists of a forum, blogs and wikis where people from different professions and different places write, discuss and contribute to eclectic topics

that range from archaeology to the Internet. It also features news about conferences and workshops on different facets of education and technology.

iTunesU
(http://www.apple.com/education/itunesu_mobile-learning/itunesu.html)

All you iPod owners out there who happen to be college students, there is something good coming up. In fact, anyone with an Apple product and with a love for learning would find Apple’s iTunesU initiative combining the best of both worlds—personal technology with education.

iTunesU uses Apple’s iTunes music store to distribute free content any time of the day, any time of the year. This service is used by hundreds of universities and schools in North America to distribute their educational content which can be then viewed over iPods, iPhones, iPod Touches and computers. In total, there are more than 30,000 audio and video files from different topics to choose from. You can even burn optical media and listen to it later at your convenience. However, you need the freely downloadable iTunes to access media from iTunesU.



One of the many courses available on iTunesU

The educational institutions that are featured on the iTunesU include Stanford, MIT, Duke, Carnegie Mellon, Lehigh, UC Berkeley and Yale, among others. It also has educational contents from other non-educational institutions like the Smithsonian Institute, The Museum of Modern Art and The New York Public Library. The content offered by these institutions vary widely in scope and depth: from a UC Berkeley philosophy course “Existentialism in Literature and Film” to “Electricity and Magnetism” by rock star Professor Walter Lewin from MIT, iTunesU has them all. They even have a series of videos on dental anatomy from University of Michigan which were recorded back in the 70s and 80s.

The USP of iTunesU is undoubtedly the ease of delivery. You can download the desired audio or video on your iPod, and listen to it whenever you want to, thus getting educated the cool way. Besides, there are no cost restrictions and you can play the media on multiple devices. In case you don’t want to keep looking at iTunes every time you look for a good pod cast, you can always go to sites like Openculture (www.oculture.com/2007/07/freeon

[linecourses.html](http://www.oculture.com/2007/07/freeonlinecourses.html)), where a list of such resources are regularly maintained.

NPTEL
(<http://www.nptel.iitm.ac.in/>)

National Programme on Technology Enhanced Learning (NPTEL) is a Government of India funded programme aimed at improving the quality of engineering college education in India. Six IITs and IISc have contributed the core material for these courses, which currently cover Civil, Computer, Electrical, Electronics and Communication and Mechanical Engineering disciplines. Information Technology and Computer Applications students and teachers would also find useful material in these video lectures. This is only the first phase of the project, with later phases aiming at including more institutes in content creation. There is content common to engineering courses like physics, chemistry, maths, management and languages.

The value of such courses is immediately apparent, when we consider the thousands of private engineering colleges that dot the length and breadth of India. Quality curriculum content development does not proceed on shoe-string budgets and needs qualified personnel, time and money. Most of these colleges can’t afford these luxuries, and hence end up with prescribing books that are suggested as part of the course by the AICTE. Students mug stuff up, pass out of the college and join the army of unemployable engineers. However, using these courses the teaching experience at IITs can be replicated to a large extent. Since these are all video courses, both students and professors can benefit from them. And they are freely available, which means no funds outlay for getting such high quality content.

Currently, the content is available in video form, with each course divided into 40 video lectures of one-hour duration each. There are a total of 129 courses with more courses being added regularly. While the inspiration behind these series of lectures is MIT’s OCW, the goals of this programme is curriculum development with focus on specific courses, unlike MIT’s primary goal of sharing content. All the lectures are put up on YouTube from where they can be downloaded and viewed later offline.

Summing Up
These are only a few of the many similar resources available on the World Wide Web. For instance, other universities have their own OCW sites which are similar to the MIT’s site mentioned here. Then again, critics point out that most of these courses are not credit earning, as you can’t get a degree after passing the tests after each course. So theoretically, you could be wasting money on Internet bills and plenty of time if you studied these courses instead of mugging up for your semester exams. On the other hand, you will probably do worse if you were to spend that time and money hanging in game parlours or restaurants, so why not give these sites a shot? You can learn plenty. ■

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